CS202T: Discrete Structures	**Quiz 1**	**7:15 PM to 8:15 PM**
Spring 2026, IIT Dharwad	**February 2, 2026**	**Total Marks: 32**

Name: ____ **Roll No.** __________________________

Instructions

- This is a question paper-cum-answer sheet, which must be submitted to the invigilator at the end of the quiz.
- Answers are to be written exclusively in the space provided after each question.
  - In your answers, state just the option number/index, and not the whole text/value per option.
  - Write your answers legibly and clearly
- Extra sheets may be used for rough work, which need not be submitted.
- No notes, books, cheat-sheet, calculator, etc. are allowed in this exam.
- Each question is of 1 mark. No negative marking.

Q1. Suppose p is the proposition "you can pass a course" and i is the proposition "you have an interest in the subject material". Express in symbols the compound proposition: "you can you pass a course only if you have an interest in the subject material."

(a) $i \to p$ (b) $p \leftrightarrow i$ (c) $p \to i$ (d) $\neg p \to \neg i$ (e) $\neg p \to i$

Ans. c

Q2. Let l be "Lois works late", let j be "John works late", and let e be "they will eat at home". Consider the proposition "If Lois or John do not work late, then they will eat at home." Which of these represents the proposition in symbols?

(a) $\neg(l \vee j) \to e$ (b) $(\neg l \wedge \neg j) \to \neg e$ (c) $\neg(l \wedge j) \to e$ (d) $e \to (\neg l \wedge \neg j)$

Ans. c

Q3. The implication $q \to \neg p$ is true for all possible assignments of truth values to p and q except for which assignment?

(a) p true, q true (b) p true, q false (c) p false, q true (d) p false, q false

Ans. a

Q4. Express in English the compound proposition $m \to \neg t$, assuming the following propositional variables.

m: "you are a member of the team" t: "you take afternoon classes."

(a) You are a member of the team only if you take afternoon classes.
(b) You are a member of the team only if you don't take afternoon classes.
(c) If you don't take afternoon classes, then you are a member of the team.
(d) If you take afternoon classes, then you are a member of the team.

Ans. b

Q5. Suppose you wish to prove this statement: If n is an integer, then $n \leq n^3$. Which of the following is correct?

(a) The given statement is true and can be proved most easily using a direct proof.
(b) The given statement is true and can be proved most easily using a proof by contraposition.
(c) The given statement is true and can be proved most easily using a proof by contradiction.
(d) The given statement is false because a counterexample can be found.

Ans. d

Q6. A number y is an additive inverse of x if $x + y = 0$. Suppose you want to prove that all real numbers have additive inverses. What quantifiers on the variables are understood?

(a) $\forall x \forall y$, where the universe for x and y is the set of all real numbers.
(b) $\exists x \forall y$, where the universe for x and y is the set of all real numbers.
(c) $\forall x \exists y$, where the universe for x and y is the set of all real numbers.
(d) $\exists x \exists y$, where the universe for x and y is the set of all real numbers.

Ans. c

Q7. Suppose you need to prove that four statements, numbered 1, 2, 3, 4, are equivalent. Which of the following lists of implications will show that the four statements are equivalent?

(a) $1 \rightarrow 2, 3 \rightarrow 1, 3 \rightarrow 2, 4 \rightarrow 2, 3 \rightarrow 4, 4 \rightarrow 3$

(b) $1 \rightarrow 2, 1 \rightarrow 3, 2 \rightarrow 4, 3 \rightarrow 4, 4 \rightarrow 1$

(c) $1 \rightarrow 2, 2 \rightarrow 1, 3 \rightarrow 4, 4 \rightarrow 3$

(d) $1 \rightarrow 3, 2 \rightarrow 1, 3 \rightarrow 2, 4 \rightarrow 1, 4 \rightarrow 2, 4 \rightarrow 3$

Ans. b

Q8. Suppose you want to prove the following statement for all integers x and y: "If x and y are even, then $3x + y$ is even." Which of these is true?

(a) This statement can be proved easily using a direct proof and it can be proved easily using a proof by contraposition.

(b) This statement can be proved easily using a direct proof, but it cannot be proved easily using a proof by contraposition.

(c) This statement can be proved easily using a proof by contraposition, but it cannot be proved easily using a direct proof.

(d) This statement cannot be proved easily using either a direct proof or a proof by contraposition.

Ans. b

Q9. Suppose you wish to prove the theorem "If $n + 1$ is an odd integer, then $n + 3$ is an odd integer." Which of these statements is correct?

(a) This theorem can be proved easily using a direct proof and it can be proved easily using a proof by contraposition.

(b) This theorem can be proved easily using a direct proof but it cannot be proved easily using a proof by contraposition.

(c) This theorem can be proved easily using a proof by contraposition but it cannot be proved easily using a direct proof.

(d) This theorem cannot be proved easily using either a direct proof or a proof by contraposition.

Ans. a

Q10. Which of these assertions is correct concerning the statement "If x^3 is irrational, then x is irrational."

(a) This statement is true, as can be shown most easily using a direct proof

(b) This statement is true, as can be shown most easily using a proof by contraposition

(c) This statement is false because a counterexample can be found

Ans. b

Q11. Which of the following is the negation of the statement "If I borrow a laptop computer, then I can complete the assignment tonight"?

(a) If I borrow a laptop computer, then I can't complete the assignment tonight.

(b) If I do not borrow a laptop computer, then I cannot complete the assignment tonight.

(c) I do not borrow a laptop computer and I do not complete the assignment tonight.

(d) I do not borrow a laptop computer and I complete the assignment tonight.

(e) I borrow a laptop computer and I do not complete the assignment tonight.

Ans. e

Q12. Which of the following is the negation of the statement "I drive to work if and only if it is rainy"?

(a) If I drive to work, then it is not rainy.

(b) I drive to work if it is not rainy.

(c) I drive to work if and only if it is not rainy.

(d) I do not drive to work if and only if it is not rainy.

(e) I do not drive to work if it is not rainy.

Ans. c

Q13. If x, y, and z are real numbers, $x < y < z$ means that x is less than y and y is less than z. Which of the following is the negation of $x < y < z$?

(a) $x \geq y$ or $y \geq z$ (b) $x \geq y \geq z$ (c) $x > y > z$ (d) $x > z$

Ans. a

Q14. Which of the following is the negation of the statement "You have meat or fish for dinner"? Assume that the disjunction "or" is exclusive.

(a) You do not have meat for dinner, and you do not have fish for dinner.
(b) Either you have both meat and fish for dinner, or you have neither.
(c) You have meat and no fish for dinner, or you have fish and no meat for dinner.

Ans. b

Q15. Which of the following statements is the negation of "Bill and Rob are absent today?"

(a) Bill or Rob is present today.
(b) Bill and Rob are present today.
(c) Either Bill is absent and Rob is present today or Bill is present and Rob is absent today.
(d) Neither Bill nor Rob are absent today.

Ans. a

Q16. Which of the following is the negation of "x is odd and y is even"?

(a) x and y are both odd
(b) x and y are both even
(c) x is odd or y is even
(d) x is even or y is odd

Ans. d

Q17. Which of the following is the negation of the statement "I will watch TV or read a book, but not both"?

(a) I will watch TV and read a book.
(b) I will neither watch TV nor read a book.
(c) Either I will watch TV and read a book, or I will do neither.

Ans. c

Q18. Which of the following is the negation of the following statement: "Everyone in the class except Lee has a laptop computer."

(a) Someone in the class other than Lee does not have a laptop computer or Lee has a laptop computer.
(b) Lee and someone else in the class have a laptop computer.
(c) Lee is the only student in the class with a laptop computer.
(d) Someone in the class other than Lee does not have a laptop computer and Lee does not have a laptop computer.

Ans. a

Q19. Which of the following is the negation of $\forall x(P(x) \to Q(x))$?

(a) $\exists x(P(x) \to Q(x))$
(b) $\exists x(P(x) \land \neg Q(x))$
(c) $\exists x(\neg P(x) \to \neg Q(x))$
(d) $\exists x(\neg P(x) \land Q(x))$

Ans. b

Q20. Suppose you want to prove this theorem by cases: "If n is an odd integer, then n^4 ends in the digit 1 or 5." What cases would you use?

(a) n ends in the digit 1 or 5
(b) n ends in one of the digits 2, 4, 6, 8, 0, or n ends in one of the digits 1, 3, 5, 7, 9
(c) n is positive, 0, or negative
(d) n ends in 1, 3, 5, 7, or 9

Ans. d

Q21. Express the following statement in symbols: "No Computer Science Major is taking any courses," using the following: $C(x)$ is the statement "x is a Computer Science Major", $T(x,y)$ is the statement "x is taking y", the universe for x is the set of all students, and the universe for y is the set of all courses.

(a) $\forall x \exists y(C(x) \to \neg T(x,y))$
(b) $\neg \exists x \exists y(C(x) \land T(x,y))$
(c) $\exists y \forall x(C(x) \land \neg T(x,y))$
(d) $\exists x \forall y(C(x) \land \neg T(x,y))$

Ans. b

Q22. Suppose you wanted to prove that the square of every even positive integer ends in 0, 4, or 6. Which type of proof would be the easiest to use?

(a) Proof by contraposition (b) Direct proof (c) Proof by cases

Ans. c

Q23. Suppose $P(x, y, z)$ is a predicate where the universe for x, y, and z is $\{1, 2\}$. Also suppose that the predicate is true in the following cases – $P(1,1,1)$, $P(1,1,2)$, $P(2,1,1)$, $P(2,2,2)$ – and false otherwise. Determine which of the following quantified statements is TRUE.

(a) $\exists x \forall y \forall z P(x, y, z)$ (b) $\forall x \exists z \forall y P(x, y, z)$ (c) $\exists x \exists y \forall z P(x, y, z)$ (d) $\exists x \exists z \forall y P(x, y, z)$

Ans. c

Q24. Which of these statements is a correct translation of the statement "There is a student in this class who does not like sushi", where $C(x)$ is the statement "x is a student in this class" and $L(x)$ is the statement "x likes sushi". Assume that the universe of discourse is the set of all people.

(a) $\exists x\big(C(x) \rightarrow \neg L\,(x)\big)$ (b) $\exists x\big(C(x) \wedge \neg L(x)\big)$ (c) $\forall x\big(C(x) \wedge \neg L(x)\big)$ (d) $\forall x\big(C(x) \rightarrow \neg L(x)\big)$

Ans. b

Q25. Which of these statements says that if a number is positive and a second number is greater than the first number, then the second number is also positive?

(a) $\forall x \exists y\big((x > 0) \rightarrow (y > 0)\big)$
(b) $\forall x \forall y([(x > 0) \wedge (y > x)] \rightarrow y > 0)$
(c) $\forall x \forall y\big((x > 0) \wedge (y > x)\big)$
(d) $\forall x \exists y\big((x > 0) \rightarrow [(y > 0) \wedge (y > x)]\big)$

Ans. b

Q26. Suppose you want to prove that the following is true for all pairs of distinct real numbers, x and y: the average of x and y lies between x and y. Which of the following can you assume, without loss of generality?

(a) x and y are even
(b) $x < 0$ and $y > 0$
(c) $x < y$
(d) x and y are integers

Ans. c

Q27. Express the following statement in symbols: "Some Computer Science courses are not being taken by any Mathematics Major," using the following: $M(x)$ is the statement "x is a Mathematics Major", $C(y)$ is the statement "y is a Computer Science course", $T(x, y)$ is the statement "x is taking y", the universe for x is the set of all students, and the universe for y is the set of all courses.

(a) $\forall y \exists x \Big(M(x) \wedge \big(C(y) \rightarrow \neg T(x, y)\big)\Big)$
(b) $\exists y \forall x \Big(C(y) \wedge \big(M(x) \rightarrow \neg T(x, y)\big)\Big)$
(c) $\exists y \exists x\big(M(x) \wedge C(y) \wedge \neg T(x, y)\big)$
(d) $\exists y \forall x \neg T\big(M(x), C(y)\big)$

Ans. b

Q28. Suppose you are examining a conjecture of the form $\forall x\big(P(x) \rightarrow Q(x)\big)$. If you are looking for a counterexample, you need to find a value x such that:

(a) $P(x)$ and $Q(x)$ are true
(b) $P(x)$ and $Q(x)$ are false
(c) $Q(x)$ is true and $P(x)$ is false
(d) $P(x)$ is true and $Q(x)$ is false

Ans. d

Q29. Which of these statements is a correct translation of the statement “Every student in this class knows the Java programming language” where $C(x)$ is the statement “x is a student in this class” and $J(x)$ is the statement “x knows Java”. Assume that the universe of discourse is the set of all people.

(a) $\exists x\big(C(x) \rightarrow J(x)\big)$
(b) $\exists x\big(C(x) \wedge J(x)\big)$
(c) $\forall x\big(C(x) \wedge J(x)\big)$
(d) $\forall x\big(C(x) \rightarrow J(x)\big)$
(e) $\forall x\big(C(x) \vee J(x)\big)$

Ans. d

Q30. Suppose you need to give a proof that there is a unique number with a certain property. Which of the following is not a valid method of doing this?

(a) Proving that there exist numbers with the desired property, and there is at most one such number.
(b) Proving that there is at least one number with the desired property, and there is at most one number with the desired property.
(c) Proving that there is a number x with the desired property, and if $y \neq x$ then y does not have the property.
(d) Proving that there is a number x with the desired property, and if y is a number with the desired property, then $y = x$
(e) Proving that there cannot be two or more numbers with the desired property.

Ans. e

Q31. Which of the following is the negation of "No one plays tennis"?

(a) Everyone plays tennis (b) Someone plays tennis (c) Someone does not play tennis

Ans. b

Q32. Which of the following is the negation of "Someone did not go to the store"?

(a) Everyone went to the store (b) No one went to the store (c) Someone went to the store

Ans. a

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CS202T: Discrete Structures	Mid-semester Exam	9:30 AM to 11:30 AM
Spring 2026, IIT Dharwad	February 26, 2026	Total Marks: 64

Name: ____ **Roll No.** __________________________

Instructions

- This is a question paper-cum-answer sheet, which must be submitted to the invigilator at the end of the exam.
- Answers are to be written exclusively in the space provided after each question.
  - In answers to objective questions, state just the option number/index, and not the whole text/value per option.
  - Write your answers legibly, clearly and concisely, especially for subjective questions
  - The limited space for answers is for a reason – articulate your thoughts, before writing!
- Extra sheets may be used for rough work, which need not be submitted.
- No notes, books, cheat-sheet, calculator, etc. are allowed in this exam.
- No negative marking.
- One or more option(s) may be correct for all MCQs. Full marks will be awarded only if all correct options are selected. No partial marks will be awarded in such questions.

Q1. [1 mark] Let $S = \{1\}$. Which of the following is an element of $P\big(P(S)\big)$ (the power set of the power set of S)?

(a) $\{\phi, \{1\}\}$ (b) $\{\phi, 1\}$ (c) $\{1\}$ (d) $\{\{\phi\}, 1\}$

Ans. a

$$P(S) = \{\phi, \{1\}\}$$
$$P(P(S)) = \{\phi, \{\phi\}, \{\{1\}\}, \{\phi, \{1\}\}\}$$

Q2. [1 mark] Let $S = \phi$. Which of the following is not a subset of $P\big(P(S)\big)$ (the power set of the power set of S)?

(a) ϕ
(b) $\{\phi\}$
(c) $\{\{\phi\}\}$
(d) $\{\phi, \{\{\phi\}\}\}$

Ans. d

$$P(S) = \{\phi\}$$
$$P(P(S)) = \{\phi, \{\phi\}\}$$

Q3. [1 mark] Let S be the set of all bit strings of length at least 1. Determine which of the following does NOT define a function $f: S \to S$. The function that takes a string as input and gives as output the

(a) reversal of the string
(b) string consisting of only the last bit in the string
(c) string 1
(d) string consisting of the first two bits of the string

Ans. d

This rule does not define a function with domain S. If s is a string of length one, $f(s)$ is not defined because s does not have two bits. For example, $f(0)$ is not defined.

Q4. [1 mark] Which of the following is true for all sets A and B?

(a) $A \cup \bar{B} = \overline{A \cap B}$
(b) $A \cup \bar{B} = (A \cap B) \cup B$
(c) $(A \cup B) \cap B = B \cup (A \cap B)$
(d) $A - (B - A) = A \cap \bar{B}$

Ans. c

Q5. [1 mark] If you need to prove that S is a proper subset of T, it is sufficient to show which of the following?

(a) $|T - S| > 0$
(b) $|S| < |T|$
(c) There is an element of T that is not an element of S
(d) $|S - T| = 0$
(e) None of these

Ans. e

Q6. [1 mark] Suppose $S = \{a, b, \{a\}\}$.

i. $\{b\} \in S$
ii. $\{a\} \subseteq P(S)$
iii. $\{a,b\} \in P(S)$
iv. $\{a,b\} \in S$
v. $\{\{a\}\} \in P(S)$
vi. $\{a,\{a\}\} \in P(S)$

Which ones of the above statements are true?

(a) (iii), (v), (vi)
(b) (ii), (iii), (vi)
(c) (i), (iii), (v)
(d) (ii), (iv), (v)
(e) (i), (ii), (vi)

Ans. a

$$P(S) = \{\phi, \{a\}, \{b\}, \{\{a\}\}, \{a,b\}, \{b,\{a\}\}, \{a,\{a\}\}, \{a,b,\{a\}\}\}$$

True: iii, v, vi
False: i, ii, iv

Q7. [1mark] Let $S = \{1,2,3,4\}$. Consider the following five statements:

i. $\{\{2\}\} \subseteq P(S)$
ii. $\{1,3\} \subseteq P(S)$
iii. $\{1,3\} \in P(S)$
iv. $\{\{2\},\{4\}\} \subseteq P(S)$
v. $\{4\} \subseteq P(S)$

Which of these five statements are true?

(a) (i), (ii), and (iv)
(b) (ii), (iv), and (v)
(c) (i), (iii), and (iv)
(d) (ii), (iii), and (iv)
(iii), (iv), and (v)

Ans. c

$$P(S) = \{\phi, \{1\}, \{2\}, \{3\}, \{4\}, \{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\}, \{1,2,3\}, \{2,3,4\}, \{1,3,4\}, \{1,2,4\}, \{1,2,3,4\}\}$$

True: i, iii, iv
False: ii, v

Q8. [1 mark] Which one of these rules defines a function f from the set of all strings of length six of letters of the alphabet to the set $\{1,2,3,4,5,6\}$?

(a) The rule that assigns to each string the number of vowels in the string. For example, $f(TAZNAV) = 2$
(b) The rule that assigns to each string the number of times Z is an element of the string. For example, $f(RZVZQC) = 2$
(c) The rule that assigns to each string the reverse of the string. For example, $f(BAQKDU) = UDKQAB$
(d) The rule that assigns to each string the position in which the first Z occurs. For example, $f(PPABZY) = 5$
(e) The rule that assigns to each string the number of distinct letters appearing in the string. For example, $f(TNVRRN) = 4$

Ans. e

This rule defines a function with the given domain and codomain. For each input string, the number of distinct letters in the string is exactly one of the integers 1, 2, 3, 4, 5, 6, all of which are elements of the given codomain.

Q9. [1 mark] Let S be the set of all bit strings of length at least 1. Determine which of the following does NOT define a function $f: S \to S$.

(a) The function that takes a string as input and gives as output the string obtained by removing all 0 bits from the string. (For example, $f(11010) = 111$.)
(b) The function that takes a string as input and gives as output the string obtained by removing the last bit from the string and placing it at the beginning of the string. (For example, $f(1110000) = 0111000$.)
(c) The function that takes a string as input and gives as output the string obtained by concatenating the string with itself. (For example, $f(0001) = 00010001$.)
(d) The function that takes a string as input and gives as output the string obtained by interchanging 0's and 1's. (For example, $f(1110) = 0001$.)1

Ans. a

This rule does not define a function from S to S. If a string s consists of only 0's, such as $s = 0000$, $f(s)$ would be the empty string, which is not in S because its length is less than one.

Q10. [1 mark] Suppose $f: R \to R$ where $f(x) = \left\lfloor \frac{3x-1}{2} \right\rfloor$. Find $\{x: f(x) = 1\}$.

(a) $\left[1, \frac{5}{3}\right)$
(b) $(0,2)$
(c) $\left[\frac{2}{3}, 3\right)$
(d) None of these

Ans. a

$$\left\lfloor \frac{3x-1}{2} \right\rfloor = 1 \Rightarrow 1 \le \frac{3x-1}{2} < 2 \Rightarrow 2 \le 3x - 1 < 4 \Rightarrow 3 \le 3x < 5 \Rightarrow 1 \le x < \frac{5}{3}$$

Q11. [1 mark] Suppose f and g are functions from $R \rightarrow R$ with $f(x) = \left\lfloor \frac{x}{2} \right\rfloor$ and $g(x) = \left\lceil \frac{5-2x}{3} \right\rceil$. Find $(f \circ g)(5)$.

(a) 1 (b) -1 (c) 0 (d) None of these

Ans. b

$g(5) = -1$ and $f(-1) = -1$

Q12. [1 mark] Suppose $f: A \rightarrow N$ is defined by the rule $f(n) = \lceil \log_2 n \rceil$. For which domain set A is the function injective?

(a) $\{3, 6, 9, 12\}$ (b) $\{2, 4, 6, 8\}$ (c) $\{1, 2, 4, 5\}$ (d) $\{1, 3, 5, 7\}$

Ans. c

$$\left.\begin{array}{r} 2^0 = 1 \\ 2^1 = 2 \\ 2^2 = 4 \\ 2^3 = 8 \end{array}\right\} \Rightarrow \left.\begin{array}{r} \log_2 1 = 0 \\ \log_2 2 = 1 \\ \log_2 4 = 2 \\ \log_2 8 = 3 \\ \log_2 16 = 4 \end{array}\right\} \Rightarrow \begin{array}{c} (-\infty, 1] \text{ maps to } 0 \\ (1,2] \text{ maps to } 1 \\ (2,4] \text{ maps to } 2 \\ (4,8] \text{ maps to } 3 \\ (8,16] \text{ maps to } 4 \end{array}$$

Q13. [1 mark] Suppose $f: R \rightarrow R$ has the rule $f(x) = \frac{1}{x^2+1}$

(a) f is injective and onto R

(b) f is injective but not onto R

(c) f is not injective but is onto R

(d) f is neither injective nor onto R

Ans. d

f is not 1-1 because, for example, $-1 \neq 1$, but $f(-1) = f(1) = \frac{1}{2}$

f is not onto R because there is no real number x such that $f(x)$ is negative.

Q14. [2 marks] Suppose we have the two propositions:

- "It is raining or I work in the yard."
- "It is not raining or I go to the library."

What conclusion can we draw from these propositions and how?

Ans.

l: I go to the library

r: It is raining

w: I work in the yard

We can use the resolution rule of inference to draw a conclusion from these propositions. In symbols the two given propositions are $(r \vee w) \wedge (\neg r \vee l)$. From resolution we have $(r \vee w) \wedge (\neg r \vee l) \rightarrow (w \vee l)$. Therefore, we can draw the conclusion "I work in the yard or I go to the library."

Q15. [2 marks] Can you tile a 17×28 checkerboard using 4×7 tiles? Explain.

Ans. Even though the number of squares on a 17×28 checkerboard is divisible by the number of squares of a 4×7 tile, the answer is NO. To see this, note that the first column of 17 squares must be covered by a combination of 4×7 tiles placed either vertically or horizontally. This means that 17 must be the sum of a multiple of 4 and a multiple of 7. However, by exhaustion, we can see that there is no solution in nonnegative integers a and b of the equation $17 = 4a + 7b$. Hence, such a tiling is impossible.

Q16. [2 marks] Find all solutions to $\lfloor x \rfloor + \lceil x \rceil = 2x$.

Ans. [1 mark to correctly analyze each of the following two cases]

$\lceil x \rceil + \lfloor x \rfloor$, the sum of two integers, must be an integer. Hence, $2x$ must be an integer, which means that either x itself is an integer, or $x + 0.5$ is an integer.

If x is an integer, then $\lceil x \rceil + \lfloor x \rfloor = x + x = 2x$

If $x + 0.5$ is an integer, then $\lceil x \rceil + \lfloor x \rfloor = (x + 0.5) + (x - 0.5) = 2x$
Thus, the solution set is $\{x | x \text{ or } x + 0.5 \text{ is an integer}\} = \left\{\frac{k}{2} | k \in \mathbf{Z}\right\}$

Q17. [2 marks] $\forall i \in Z^+, A_i = \left[0, \frac{1}{i}\right)$. If $n \in Z^+$, find the union and intersection of these sets from $i = 1$ to n.

Ans. [1 mark each for union and intersection]
Observe that each A_i is a subset of all previous $A_j, \forall j < i$

$$\bigcup_{i=1}^{n} A_i = [0,1) = A_1$$
$$\bigcap_{i=1}^{n} A_i = \left[0, \frac{1}{n}\right) = A_n$$

Q18. [3 marks] Suppose that A and B are sets such that $P(A \cup B) \subseteq P(A) \cup P(B)$. Prove that $A \subseteq B$ or $B \subseteq A$.

Ans. Proof by contraposition, i.e. we pursue a direct proof of the following:
If $A \subseteq B$ or $B \subseteq A$ are false, then $P(A \cup B) \subseteq P(A) \cup P(B)$ is false.

$A \subseteq B$ or $B \subseteq A$ are false $\Rightarrow \exists a \in A - B$ and $\exists b \in B - A$

Consider the set $S = \{a, b\}$
$\Rightarrow S \subseteq A \cup B$ and S is neither subset of A nor that of B
$\Rightarrow S \in P(A \cup B), S \notin P(A), S \notin P(B)$
$\Rightarrow S \in P(A \cup B), S \notin P(A) \cup P(B)$
$\Rightarrow P(A \cup B) \subseteq P(A) \cup P(B)$ is false

Q19. [4 marks] Suppose a and b are integers such that $2a = b^2 + 3$. Prove that a is the sum of three squares.

Ans. Note that $2a$ is even $\Rightarrow b^2 + 3$ is even $\Rightarrow b^2$ is odd $\Rightarrow b$ is odd $\Rightarrow b = 2n + 1$, for some integer n
$\Rightarrow 2a = (2n + 1)^2 + 3 = 4n^2 + 4n + 4$
$\Rightarrow a = 2n^2 + 2n + 2 = n^2 + (n + 1)^2 + 1^2$

Q20. [4 marks] You meet two people, A and B. Each person either always tells the truth (i.e., the person is a knight) or always lies (i.e., the person is a knave). Person A tells you, "We are not both truthtellers." Determine, if possible, which type of person each one is.

Ans.
One way to solve the problem is by considering each of the four possible cases: both lie, both tell the truth, A lies and B tells the truth, A tells the truth and B lies. We examine each of the four cases separately.
Both lie: That would mean that neither one is a truthteller. The statement "We are not both truthtellers" is true because they are not both truthtellers. Thus, A is telling the truth, which contradicts the assumption that both are liars. Therefore, this case cannot happen.
Both tell the truth: Therefore, the statement "We are not both truthtellers" is a lie. Thus, A is a liar, contradicting the assumption that both are truthtellers. Therefore, this case cannot happen.
A lies and B tells the truth: If A is a liar, A's statement "We are not both truthtellers" must be a lie. Therefore, A and B must both be truthtellers. But this contradicts the assumption that A is a liar. Therefore, this case cannot happen.
A tells the truth and B lies: In this case A's statement "We are not both truthtellers" is true and no contradiction is obtained. Therefore, this case is not ruled out.
Therefore, the only possibility is that A is a truthteller and B is a liar.

Another way to solve the problem is to see where the assumption "A is a liar" leads us. If we assume that A always lies, then A's statement "We are not both truthtellers" must be false. Therefore, both A and B must be truthtellers. Therefore, A's statement "We are not both truthtellers" must be true, a contradiction of the fact that the statement is false.
This says that A is not a liar. Therefore, A is a truthteller. Therefore, A's statement "We are not both truthtellers" must be true. Because A is a truthteller, A's statement forces B to be a liar.

Q21. [4 marks] Determine whether this argument is valid:

- Lynn works part time or full time.
- If Lynn does not play on the team, then she does not work part time.
- If Lynn plays on the team, she is busy.
- Lynn does not work full time.
- Therefore, Lynn is busy.

Ans. Using the following variables,

p: Lynn works part time
f: Lynn works full time
t: Lynn plays on the team
b: Lynn is busy

the argument can be written in symbols as:

$$\begin{array}{c} p \vee f \\ \neg t \rightarrow \neg p \\ t \rightarrow b \\ \neg f \\ \hline \therefore b \end{array}$$

Rules of logic can be used to give a proof that the above argument is valid. We begin with the four hypotheses and show how to derive the conclusion, b

1. $p \vee f$ premise
2. $\neg t \rightarrow \neg p$ premise
3. $t \rightarrow b$ premise
4. $\neg f$ premise
5. p disjunctive syllogism on (1) and (4)
6. $p \rightarrow t$ contrapositive of (2)
7. t modus ponens on (5) and (6)
8. b modus ponens on (7) and (3)

Alternatively, truth table can also be used to verify the argument as follows.

p	f	t	b	$p \wedge f$	$\neg t \rightarrow \neg p$	$t \rightarrow b$	$\neg f$	b
T	T	T	T	T	T	T	F	T
T	T	T	F	T	T	F	F	F
T	T	F	T	T	F	T	F	T
T	T	F	F	T	F	T	F	F
T	F	T	T	T	T	T	T	T
T	F	T	F	T	T	F	T	F
T	F	F	T	T	F	T	T	T
T	F	F	F	T	F	T	T	F
F	T	T	T	T	T	T	F	T
F	T	T	F	T	T	F	F	F
F	T	F	T	T	T	T	F	T
F	T	F	F	T	T	T	F	F
F	F	T	T	F	T	T	T	T
F	F	T	F	F	T	F	T	F
F	F	F	T	F	T	T	T	T
F	F	F	F	F	T	T	T	F

We need to examine all cases where the hypotheses (columns 5, 6, 7, 8) are all true. There is only one case in which all four hypotheses are true (row 5), and in this case the conclusion is also true. Therefore, the argument is valid.

Q22. [4 marks] Suppose a matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with real numbers as entries commutes under multiplication with all real 2×2 matrices, i.e. $AB = BA, \forall 2 \times 2$ matrices B with real numbers as entries. What form must A have?

Ans. [2 marks each for any two of the following three observations]

1. $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow c = 0, a = d$
2. $B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \Rightarrow b = 0, a = d$
3. $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ or $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow b = c = 0$

Any two of the above observations leads to the conclusion that $A = \lambda I$, where I is the Identity matrix and $\lambda \in \mathbf{R}$

Q23. [4 marks] Show that there is no one-to-one correspondence from the set of positive integers to the power set of the set of positive integers.

Ans.

Approach 1

[1 mark] Represent a subset of the set of positive integers as an infinite bit string with i^{th} bit as 1 if i belongs to the subset and 0 otherwise.

[1 mark] Suppose that you can list these infinite strings in a sequence indexed by the positive integers.

[1 mark] Construct a new bit string with its i^{th} bit equal to the complement of the i^{th} bit of the i^{th} string in the list.

[1 mark] Show that this new bit string cannot appear in the list.

Approach 2

Assume, for contradiction, that $f: Z^+ \to P(Z^+)$ is a bijection

[1 mark] Define $D = \{n \in Z^+ | n \notin f(n)\} \Rightarrow D \subseteq Z^+ \Rightarrow D \in P(Z^+)$

[1 mark] Since f is onto, there exists $k \in Z^+$ such that $f(k) = D$

[1 mark] *Case 1*: $k \in D$

$$k \in D \xrightarrow{\text{By definition of } D} k \notin f(k) \xrightarrow{\because f(k)=D} k \notin D \to \text{Contradiction}$$

[1 mark] *Case 2*: $k \notin D$

$$k \notin D \xrightarrow{\text{By definition of } D} k \in f(k) \xrightarrow{\because f(k)=D} k \in D \to \text{Contradiction}$$

Hence no such bijection exists.

Q24. [5 marks] Find a recurrence relation for the number of strings of letters of the ordinary alphabet that do not have adjacent vowels.

Ans. Let

$$v_n = \text{Number of such strings of length } n \text{ ending in a vowel}$$
$$c_n = \text{Number of such strings of length } n \text{ ending in a consonant}$$

[1 mark] Then the total number of valid strings is

$$a_n = v_n + c_n$$

[1 mark] *__Strings ending in a vowel__*. The previous letter must be a consonant (to prevent adjacency), and there are 5 choices for the vowel.

$$v_n = 5c_{n-1}$$

[1 mark] *__Strings ending in a consonant__*. The previous letter can be any valid letter, and there are 21 choices for the consonant.

$$c_n = 21a_{n-1}$$

Combining the above three equations:

$$a_n = v_n + c_n = 5c_{n-1} + 21a_{n-1}$$

But

$$c_{n-1} = 21a_{n-2}$$

Substituting we have:

$$a_n = 5(21a_{n-2}) + 21a_{n-1}$$
$$a_n = 21a_{n-1} + 105a_{n-2}, \forall n \geq 2$$

[2 marks] Initial conditions:

$$a_0 = 1 \text{ (empty string)}$$
$$a_1 = 26 \text{ (single letter)}$$
$$\boxed{a_n = 21a_{n-1} + 105a_{n-2}, a_0 = 1,\ a_1 = 26}$$

Q25. [7 marks] Use the Principle of Mathematical Induction to show that the following inequality is true for all integers $n \geq 2$.

$$\sum_{i=1}^{n} \frac{1}{\sqrt{i}} > \sqrt{n}$$

Ans.

[1 mark] **Base case $n = 2$:**

$$1 + \frac{1}{\sqrt{2}} > \sqrt{2}$$

[1 mark] **Inductive hypothesis:** Assume

$$\sum_{i=1}^{n} \frac{1}{\sqrt{i}} > \sqrt{n}.$$

[5 marks] **Inductive step**:

$$\sum_{i=1}^{n+1} \frac{1}{\sqrt{i}} = \sum_{i=1}^{n} \frac{1}{\sqrt{i}} + \frac{1}{\sqrt{n+1}}$$

Using the hypothesis:

$$> \sqrt{n} + \frac{1}{\sqrt{n+1}}$$

Since

$$\sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

and

$$\frac{1}{\sqrt{n+1}} > \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

we obtain

$$\sum_{i=1}^{n+1} \frac{1}{\sqrt{i}} > \sqrt{n+1}$$

Thus, the inequality holds for all $n \geq 2$:

$$\sum_{i=1}^{n} \frac{1}{\sqrt{i}} > \sqrt{n}$$

Q26. [8 marks] Prove that $\neg\big[r \vee \big(q \wedge (\neg r \to \neg p)\big)\big] \equiv \neg r \wedge (p \vee \neg q)$ by using a

(a) truth table
(b) series of logical equivalences

Ans.

(a) Truth table [4 marks]

We construct the truth tables for $\neg[r \vee (q \wedge (\neg r \to \neg p))]$ and for $\equiv \neg r \wedge (p \vee \neg q)$, and show that they are identical. We insert "intermediate" columns as we build each of the propositions.

p	q	r	$\neg p$	$\neg r$	$\neg r \to \neg p$	$q \wedge (\neg r \to \neg p)$	$r \vee (q \wedge (\neg r \to \neg p))$	$\neg(r \vee (q \wedge (\neg r \to \neg p)))$	$p \vee \neg q$	$\neg r \wedge (p \vee \neg q)$
T	T	T	F	F	T	T	T	F	T	F
T	T	F	F	T	F	F	F	T	T	T
T	F	T	F	F	T	F	T	F	T	F
T	F	F	F	T	F	F	F	T	T	T
F	T	T	T	F	T	T	T	F	F	F
F	T	F	T	T	T	T	T	F	F	F
F	F	T	T	F	T	F	T	F	T	F
F	F	F	T	T	T	F	F	T	T	T

Note that the ninth and eleventh columns are identical. Therefore, the two propositions are equivalent.

(b) series of logical equivalences [4 marks]

We will begin with $\neg[r \vee (q \wedge (\neg r \to \neg p))]$ and use rules of logic to show that this is equivalent to $\neg r \wedge (p \vee \neg q)$. Here is one possible proof:

$$
\begin{aligned}
\neg[r \vee (q \wedge (\neg r \to \neg p))] &\equiv \neg r \wedge \neg(q \wedge (\neg r \to \neg p)) && \text{De Morgan's law} \\
&\equiv \neg r \wedge \neg(q \wedge (\neg\neg r \vee \neg p)) && \text{conditional rewritten as disjunction} \\
&\equiv \neg r \wedge \neg(q \wedge (r \vee \neg p)) && \text{double negation law} \\
&\equiv \neg r \wedge (\neg q \vee \neg(r \vee \neg p)) && \text{De Morgan's law} \\
&\equiv \neg r \wedge (\neg q \vee (\neg r \wedge p)) && \text{De Morgan's law and double negation} \\
&\equiv (\neg r \wedge \neg q) \vee (\neg r \wedge (\neg r \wedge p)) && \text{distributive law} \\
&\equiv (\neg r \wedge \neg q) \vee ((\neg r \wedge \neg r) \wedge p) && \text{associative law} \\
&\equiv (\neg r \wedge \neg q) \vee (\neg r \wedge p) && \text{idempotent law} \\
&\equiv \neg r \wedge (\neg q \vee p) && \text{distributive law} \\
&\equiv \neg r \wedge (p \vee \neg q) && \text{commutative law}
\end{aligned}
$$

CS202T: Discrete Structures	**Quiz #2**	**7:30 PM to 8:30 PM**
Spring 2026, IIT Dharwad	**March 26, 2026**	**Total Marks: 36**

Name: ____ **Roll No.** __________________________

Instructions

- This is a question paper-cum-answer sheet, which must be submitted to the invigilator at the end of the exam.
- Answers are to be written exclusively in the space provided after each question.
- In your answers, state just the option number/index, and not the whole text/value per option.
- Extra sheets may be used for rough work, but need not be submitted.
- No notes, books, cheat-sheet, calculator, etc. are allowed in this exam.
- Answers written using pencil will be disqualified.
- Marking scheme
    - 1 mark for each correct answer
    - -0.25 mark for each incorrect answer

Q1. In a Strong Induction proof for a sequence defined by $a_n = a_{n-1} + a_{n-3}$, how many base cases are strictly required to begin the inductive step?

(a) 1
(b) 2
(c) 3
(d) It depends on whether n starts at 0 or 1.

Ans. c

Q2. What is the "Inductive Hypothesis" in a Strong Induction proof for $P(n)$ starting at $n = 0$?

(a) $P(k)$ is true
(b) $P(k+1)$ is true
(c) $\forall i \in \{0,1,2, \dots k\}, P(i)$ is true
(d) $P(0) \wedge P(k)$ is true

Ans. c

Q3. In a recursive definition of a set S, if the base case is $3 \in S$ and the rule is $x, y \in S \rightarrow (x + y) \in S$, what property can be proven about all $n \in S$ via structural induction?

(a) n is an odd number
(b) n is a multiple of 3
(c) n is a power of 3
(d) n is prime

Ans. b

Q4. Which principle serves as the foundational "floor" that allows any form of induction on the natural numbers to work?

(a) Law of Large Numbers
(b) Well-Ordering Principle
(c) Axiom of Choice
(d) Pigeonhole Principle

Ans. b

Q5. In structural induction, what is the primary goal of the recursive step?

(a) To prove that if the statement is true for elements used to construct new ones, it holds for those new elements.
(b) To solve the characteristic equation of the recursively defined set.
(c) To show the result holds for elements specified in the basis step of the definition.
(d) To demonstrate that the set has the well-ordering property.

Ans. a

Q6. Which of the following is a valid recursive definition of a function $f: \mathbb{N} \rightarrow \mathbb{Z}$?

(a) $f(0) = 0$ and $f(n) = \frac{3}{f(n-1)}, \forall n \geq 1$
(b) $f(0) = 1, f(1) = 1$ and $f(n) = f(n-1) - 3f(n-2), \forall n \geq 2$
(c) $f(0) = 2, f(1) = 0$ and $f(n) = 5 + f(n-1), \forall n \geq 1$
(d) $f(0) = 1$ and $f(n) = 3f(n-2), \forall n \geq 1$

Ans. b

Q7. Which condition defines the lexicographic ordering on the set of ordered pairs of nonnegative integers $\mathbb{N} \times \mathbb{N}$?

(a) $(x_1, y_1) \leq (x_2, y_2)$ if $x_1 + y_1 \leq x_2 + y_2$
(b) $(x_1, y_1) \leq (x_2, y_2)$ if $y1 < y2$ regardless of the values of x_1 and x_2
(c) $(x_1, y_1) \leq (x_2, y_2)$ if $x_1 < x_2$ or $(x_1 = x_2$ and $y_1 < y_2)$
(d) $(x_1, y_1) \leq (x_2, y_2)$ only if both $x_1 \leq x_2$ and $y_1 \leq y_2$

Ans. c

Q8. Give a recursive definition for the sequence 1, 2, 3, 1, 2, 3, 1, 2, 3, ..., assuming that the pattern "1, 2, 3" continues.

(a) $a_n = (a_{n-1} + 1) \bmod 4, a_0 = 1$
(b) $a_n = (a_{n-1} + 1) \bmod 3, a_0 = 1$
(c) $a_n = a_{n-1} \bmod 3 + 1, a_0 = 1$
(d) $a_n = (a_{n-1} + a_{n-2}) \bmod 3 + 1, a_0 = 1, a_1 = 2$

Ans. c

Q9. If $f: \mathbb{N} \to \mathbb{R}$ is defined recursively by $f(0) = 2, f(1) = 5, f(n) = 3f(n-1) - 2f(n-2)$ then $f(4)$ is equal to which of the following?

(a) 18 (b) 47 (c) 23 (d) 95 (e) None of these

Ans. b

Q10. Which of the following is a recursive definition for $a_n = 4n + 3$?

(a) $a_n = 2a_{n-1} + 1, a_0 = 3$
(b) $a_n = a_{n-1} + 4n, a_0 = 3$
(c) $a_n = a_{n-2} + 8, a_0 = 3$
(d) $a_n = 2a_{n-1} - a_{n-2}, a_0 = 3, a_1 = 7$

Ans. d

Q11. You begin with a square with side length 1. At the end of minute 1 you increase the side length by 1. You continue to increase the side length by 1 at the end of each minute. Let a_n equal the area of the square at the end of minute n, immediately after you increase its side length. Which of the following is a recurrence relation for a_n?

(a) $a_n = a_{n-1} + (2n + 1)$
(b) $a_n = a_{n-1} + (n - 1)^2$
(c) $a_n = a_{n-1}^2$
(d) $a_n = (n + 1)^2$

Ans. a

Q12. For which of the following is the recursively defined set S equal to $\{a, 2a, 3a, 4a, 5a, \ldots\}$?

(a) $a \in S; x \in S \to x + x \in S$
(b) $a \in S; x \in S \to x^2 \in S$
(c) $a \in S; x \in S \to 2x \in S$
(d) $a \in S; x \in S \to a + x \in S$

Ans. d

Q13. For which of the following is the set S equal to the set of all positive integers not divisible by 3?

(a) $1 \in S; x \in S \to 2x + 2 \in S$
(b) $1 \in S; x \in S \to x + 3 \in S$
(c) $1 \in S; 2 \in S; x \in S \to x + 3 \in S$
(d) $1 \in S; x \in S \to 3x + 1 \in S$

Ans. c

Q14. For which of the following is the set S equal to $\{(a, b): a, b \text{ nonnegative integers}\}$?

(a) $(0,0) \in S; (x, y) \in S \to (x + 1, y) \in S$ and $(x, y + 1) \in S$
(b) $(0,0) \in S; (x, y) \in S \to (x + 1, y + 1) \in S$
(c) $(0,0) \in S; (x, y) \in S \to (x + y, y) \in S$ and $(x, x + y) \in S$
(d) None of these

Ans. a

Q15. Which of the following is a recursive definition of the function $f: \mathbb{N} \to \mathbb{Z}$, where $f(n) = n^2$?

(a) $f(n) = nf(n-1) + 1, f(0) = 0$
(b) $f(n) = f(n-1) + (2n - 1), f(0) = 0$
(c) $f(n) = f(n-1)^2, f(0) = 0$
(d) $f(n) = f(n-1) + (2n + 1), f(0) = 0$
(e) $f(n) = 2f(n-1) + 2$

Ans. b

Q16. Suppose you are hired by a company at an initial salary of ₹30,000. At the end of each year, you receive a 3% raise, and then an additional ₹1000 increase. Let a_n equal your salary at the end of n years with the company. Find a recurrence relation for a_n.

(a) $a_n = 1.03a_{n-1} + 1000$
(b) $a_n = (1000 + 0.03)a_{n-1}$
(c) $a_n = 1.03(a_{n-1} + 1000)$
(d) None of these

Ans. a

Q17. Suppose you begin with one pair of newborn rabbits. At the end of the third month, and at the end of every month thereafter, they give birth to two pairs of rabbits. Each pair of offspring reproduces according to the same rule. Assume that none of the rabbits die. Let f_n be the number of pairs of rabbits at the end of month n, just after the new pairs have been born. We have $f_1 = 1$, $f_2 = 1$, and $f_3 = 3$. Which of the following is a recurrence relation for f?

(a) $f_n = f_{n-1} + 2f_{n-3}$
(b) $f_n = f_{n-1} + f_{n-3}$
(c) $f_n = f_{n-1} + 2f_{n-2}$
(d) $f_n = f_{n-1} + f_{n-2} + f_{n-3}$

Ans. a

Q18. A dormitory has 40 students 12 sophomores, 8 juniors, and 20 seniors. Which of the following is equal to the number of ways to put all 40 in a row for a picture, with all 12 sophomores on the left, all 8 juniors in the middle, and all 20 seniors on the right?

(a) $C(12,12) \cdot C(8,8) \cdot C(20,20)$
(b) $12!\,8!\,20!$
(c) $\frac{40!}{3}$
(d) $\frac{40!}{12!8!20!}$

Ans. b

Q19. Suppose $f(n)$ has the recursive definition $f(n) = \frac{3}{f(n-1)} + 1$ and you know that $f(3) = 7$. What is the value $f(1)$?

(a) $\frac{2}{3}$
(b) $\frac{1}{2}$
(c) -6
(d) $-\frac{6}{5}$

Ans. c

Q20. For which of the following is the recursively defined set S equal to the set of odd positive integers?

(a) $1 \in S; 3 \in S; x \in S \rightarrow x + 4 \in S$
(b) $99 \in S; x \in S \rightarrow x - 2 \in S$
(c) $2 \in S; x \in S \rightarrow x + 2 \in S$
(d) $1 \in S; x \in S \rightarrow 2x + 1 \in S$
(e) None of these

Ans. a

Q21. Assume that you have an ordinary deck of 52 playing cards. How many possible 7-card poker hands are there that contain at least one face card (Jack, Queen, or King)?

(a) $C(52,7) \cdot C(40,7)$
(b) $C(12,1) \cdot C(40,6)$
(c) $C(4,1) \cdot C(4,1) \cdot C(4,1) \cdot C(40,4)$
(d) $C(12,3) \cdot C(40,4)$

Ans. This question is disqualified, on account of NO correct options.

Q22. How many different passwords are possible if each password consists of six characters where each character is either an uppercase letter, a lowercase letter, or a digit, and at least one digit must be included in the password?

(a) $62^6 - 10^6$
(b) $62^6 - 26^6$
(c) $62^6 - 36^6$
(d) $62^6 - 52^6$

Ans. d

Q23. The price of a stock is initially ₹50. Suppose that each morning the stock price increases by 2% and then in the afternoon the price decreases by 2%. Let a_n be the price of the stock at the end of day n. Which of the following is a recurrence relation for a_n?

(a) $a_n = (0.98 \times 1.02)a_{n-1}$
(b) $a_n = a_{n-1}$
(c) $a_n = 1.02a_{n-1} - 0.02a_{n-1}$
(d) $a_n = 1.02a_{n-1} \times 0.98a_{n-1}$

Ans. a

Q24. How many bit strings (strings of 0's and 1's) are there of length 7 that have more 0's than 1's?

(a) 2^6
(b) $2^4 + 2^5 + 2^6 + 2^7$
(c) $C(7,4) \cdot C(7,5) \cdot C(7,6) \cdot C(7,7)$
(d) $2 \cdot 2 \cdot 2 \cdot 1 \cdot 1 \cdot 1 \cdot 1$

Ans. a

Q25. If $a_n = a_{n-1}^2$ and $a_0 = 2$, then a_n is equal to which of the following?

(a) 2^{2^n} (b) 2^{n+1} (c) $(2^n)^2$ (d) 2^{2n}

Ans. a

Q26. Suppose $a_n = a_{n-1} + 2^n$. Then a_n is equal to which of the following?

(a) $a_{n-2} + 2^n + 2^{n-1}$ (b) $a_{n-2} + 2^{n+1}$ (c) $a_{n-2} + 2^n$ (d) $a_{n-2} + 2^{n-1}$

Ans. a

Q27. The population of the United States in 2006 is approximately 298,000,000 and is growing at a rate of 1.2% per year. Let a_n be the population n years from 2006. If you want to find how long it will take for the population to double, you must find a subscript n such that:

(a) $a_n = 2a_{n-1}$ (b) $2a_0 = a_n$ (c) $a_n = 2a_{2006}$ (d) $a_{2n} = a_n$

Ans. b

Q28. Suppose a sequence $a_1, a_2, a_3, \ldots$ begins 3, 7, 3, 7, 3, 7, ..., and this pattern continues. Which of the following is a formula for this sequence?

(a) $a_n = 5 + (-2)^n$ (b) $a_n = 5 + 2(-1)^{n-1}$ (c) $a_n = 3 + 4^{n-1}$ (d) $a_n = a_{n-2}$

Ans. This question is disqualified, on account of NO correct options.

Q29. Which of the following is a recurrence relation for the sequence that begins 3, 6, 9, 12, 15,...?

(a) $a_n = a_{n-2} + 6$ (b) $a_n = 3n$ (c) $a_n = 3a_{n-1}$ (d) $a_n = a_{n-1} + 3n$

Ans. a

Q30. Which of the following is a formula for the sequence 3, 6, 12, 24, 48, ...? Assume that the first term in the sequence is called a_0.

(a) $a_n = 3 \cdot 2^{n-1}$ (b) $a_n = 3n + 3$ (c) $a_n = 2a_{n-1}$ (d) $a_n = 3(n+1)$ (e) None of these

Ans. c OR e OR both c, e

Q31. How many strings of all the first seven letters of the alphabet (A, B, C, D, E, F, G) are there that contain no repeated letters and begin or end with a vowel (A or E)?

(a) $7! - 5!$
(b) $2 \cdot 6! + 2 \cdot 6!$
(c) $2 \cdot 6! + 2 \cdot 6! - 2 \cdot 5!$
(d) $6! + 6! - 5!$
(e) $6! + 6!$

Ans. c
Strings which begin with a vowel + Strings which end with a vowel – Strings which begin and end with a vowel

Q32. You have 12 balls, numbered 1 through 12, which you want to place into 4 boxes, numbered 1 through 4. If boxes can remain empty, in how many ways can the 12 balls be distributed among the 4 boxes?

(a) 4^{12} (b) 12^4 (c) $C(12,4)$ (d) $P(12,4)$ (e) $\frac{12!}{4!}$

Ans. a
Each ball can be placed in any of the four boxes (i.e. four choices)

Q33. How many different license plates are available if the license plate pattern consists of three letters that cannot be repeated followed by three digits that can be repeated? (Assume that all letters are uppercase and digits are 0, 1, ..., 9.)

(a) $26^3 \cdot 10^3$
(b) $C(26,3) \cdot 10^3$
(c) $26 \cdot 25 \cdot 24 \cdot 10^3$
(d) $C(26,3) \cdot C(10,3)$
(e) $26 \cdot 25 \cdot 24 \cdot 10 \cdot 1 \cdot 1$

Ans. c

Q34. A class consists of 12 women and 10 men. How many ways are there to form a committee of size six if the committee has more women than men?

(a) $C(12,4)C(10,2)$
(b) $\left(C(12,4)\cdot C(10,2)\right)\cdot\left(C(12,5)\cdot C(10,1)\right)\cdot\left(C(12,6)\cdot C(10,0)\right)$
(c) $C(12,4)\cdot C(10,2)+C(12,5)\cdot C(10,1)+C(12,6)\cdot C(10,0)$
(d) $C(22,6)-[C(10,6)C(12,0)-C(10,5)\cdot C(12,1)-C(10,4)C(12,2)]$

Ans. c
Use the sum rule to count the committees of four women and three men OR five women and one man OR six women and no men

Q35. How many non-negative integer solutions exist for the inequality $x_1+x_2+x_3\leq 10$?

(a) $\binom{12}{2}$ (b) $\binom{13}{3}$ (c) $\binom{10}{3}$ (d) $\binom{14}{4}$

Ans. b
This looks like a standard stars and bars problem, but the "less than or equal to" sign is tricky. Try introducing a "dummy variable" (x_4) to represent the leftover amount needed to reach exactly 10.

Q36. Let $a_n \stackrel{\text{def}}{=}$ the number of bit strings of length n with two consecutive 0's. Which of the following is a recurrence relation for a_n?

(a) $a_n=a_{n-1}+a_{n-2}+2^{n-2}$
(b) $a_n=2a_{n-1}+2a_{n-2}$
(c) $a_n=2a_{n-1}+2a_{n-2}-1$
(d) $a_n=a_{n-1}+a_{n-2}+2^n$
(e) None of these

Ans. a
$a_n=$ Number of bit strings starting with 1 + Number of bit strings starting with 01 + Number of bit strings starting with $00=a_{n-1}+a_{n-2}+2^{n-2}$

CS202T: Discrete Structures Spring 2026, IIT Dharwad	End-sem Exam April 29, 2026	9:30 AM to 12:30 PM Total Marks: 78

Name: ____ **Roll No.** __________________________

Instructions

- This is a question paper-cum-answer sheet, which must be submitted to the invigilator at the end of the exam.
  - Extra sheets may be used for rough work, and must be SEPARATELY returned to the invigilator
  - Extra sheets used for rough work must NOT be attached to the question paper-cum-answer sheet
- Answers are to be written exclusively in the space provided after each question.
  - In objective questions, state just the option number/index, and not the whole text/value per option.
  - Write your answers legibly, clearly, and concisely, especially for subjective questions
  - The limited space for answers is for a reason – articulate your thoughts, before writing!
- No notes, books, cheat-sheet, calculator, etc. are allowed in this exam.
- Answers written using a pencil will be disqualified.
- Negative marking of 0.25 marks is applicable for each incorrectly answered MCQ question.

Q1. [1 mark] Suppose you are trying to prove a theorem about all rational numbers a/b. Which of the following can be assumed without loss of generality?

(a) a/b has been reduced to lowest terms
(b) $a < b$
(c) $a/b \geq 0$
(d) a and b are odd integers
(e) a/b can be written as a terminating decimal

Ans. a

Q2. [1 mark] Suppose you are examining a conjecture of the form $\forall x(P(x) \wedge Q(x))$. To show that the conjecture is false, you MUST show which of the following?

(a) There is a value x_1 such that $P(x_1)$ is false and a value x_2 such that $Q(x_2)$ is false.
(b) There is a value x such that either $P(x)$ is false or $Q(x)$ is false.
(c) For every choice of x, $P(x)$ and $Q(x)$ are both false.
(d) For every choice of x, either $P(x)$ is false or $Q(x)$ is false.

Ans. b

Q3. [1 mark] Express the following statement in symbols: "Every Mathematics Major is taking a Computer Science course," using the following: $M(x)$ is the statement "x is a Mathematics Major", $C(y)$ is the statement "y is a Computer Science course", $T(x,y)$ is the statement "x is taking y", the universe for x is the set of all students, and the universe for y is the set of all courses.

(a) $\forall x \exists y[M(x) \rightarrow (C(y) \wedge T(x,y))]$
(b) $\forall x \exists y[M(x) \wedge C(y) \wedge T(x,y)]$
(c) $\forall x \exists y[M(x) \rightarrow T(x, C(y))]$
(d) $\exists y \forall x[M(x) \rightarrow (T(x,y) \wedge C(y))]$
(e) $\forall y \exists x(M(x) \wedge C(y) \wedge T(x,y))$

Ans. a

Q4. [1 mark] Suppose $P(x,y)$ is a predicate where the universe for x and y is $\{1,2,3\}$. Also suppose that the predicate is true in the following cases – $P(1,2)$, $P(2,1)$, $P(2,2)$, $P(2,3)$, $P(3,1)$, $P(3,2)$ – and false otherwise. Determine which of the following quantified statements is FALSE.

(a) $\forall y \exists x \neg P(x,y)$
(b) $\exists y \forall x P(x,y)$
(c) $\forall x \exists y(x \neq y \wedge P(x,y))$
(d) $\exists x \forall y P(x,y)$

Ans. a

Q5. [1 mark] Suppose h and c are these propositions: h: "I go hiking" c: "it is a cold day." Express in symbols the compound proposition "I don't go hiking when it is a cold day."

(a) $h \rightarrow c$ (b) $c \rightarrow \neg h$ (c) $\neg c \rightarrow h$ (d) $\neg h \rightarrow c$

Ans. b

Q6. [1 mark] Which of these statements says that "Every number has exactly one additive inverse."? Assume that the universe for all variables consists of all real numbers.

(a) $\forall x \exists y \forall z[(x+y=0) \land ((x+z=0) \to (y=z))]$
(b) $\forall x \forall y \exists z(x+y=x+z=0)$
(c) $\forall x \exists y(x+y=0)$
(d) $\forall x \exists y \exists z[(x+y=0) \land (x+z=0)]$

Ans. a

Q7. [1 mark] A class consists of 12 women and 10 men. How many ways are there to form a committee of size six if the committee has equal numbers of women and men?

(a) $C(12,3)+C(10,3)$
(b) $C(12,3) \cdot C(10,3)$
(c) $C(22,6)$
(d) $C(22,6)-C(12,6)-C(10,6)$

Ans. b

Q8. [1 mark] Which of the following statement is the negation of "Bill or Rob is absent today?"

(a) Bill or Rob is present today.
(b) Bill and Rob are present today.
(c) Either Bill is absent and Rob is present today or Bill is absent and Rob is present today.
(d) Neither Bill nor Rob is present today.

Ans. b

Q9. [1 mark] Express in English a compound proposition logically equivalent to $w \to \neg r$, assuming the following.

r: "it is rainy" w: "it is windy."

(a) If it is rainy, then it is not windy.
(b) It is windy, but not rainy.
(c) Whenever it is not windy, it is rainy.
(d) Being windy is sufficient for it to be rainy.

Ans. a

Q10. [1 mark] Suppose $a_n = 2a_{n-1} + 2n + 1$. Then a_{n-1} is equal to which of the following?

(a) $2a_{n-2}+2n-1$ (b) $2a_{n-2}+2n$ (c) $2a_{n-2}+2n+1$ (d) None of these

Ans. a

Q11. [1 mark] The population of the United States in 2006 is approximately 298,000,000 and is growing at a rate of 1.2% per year. Let a_n be the population n years from 2006. Which of the following is a recurrence relation for a_n?

(a) $a_n = a_{n-1} + 0.012a_{n-1}$
(b) $a_n = 1.2a_{n-1}$
(c) $a_n = 1.012a_{n-1} + 298{,}000{,}000$
(d) $a_n = 0.12a_{n-1} + 2006$
(e) $a_n = 0.012a_{n-1}$
(f) $a_n = 298{,}000{,}000a_{n-1}$

Ans. a

Q12. [1 mark] Suppose inflation decreases the value of money by 3% per year. Which formula describes $a_n \overset{\text{def}}{=}$ the value (in rupees) of ₹1000 after n years?

(a) $a_n = (0.97)^n 1000$ (b) $a_n = n(0.97)1000$ (c) $a_n = -(0.03)1000$ (d) $a_n = 1000(0.03)^n$

Ans. a

Q13. [1 mark] Suppose a sequence a_1, a_2, a_3, ... begins 3, 7, 3, 7, 3, 7, ..., and this pattern continues. Which of the following is a recurrence relation for this sequence?

(a) $a_n = a_{n-1} \pm 4$
(b) $a_n = a_{n-1} + 4$
(c) $a_n = a_{n-1} + (-2)^n$
(d) $a_n = a_{n-1} + 4(-1)^n$
(e) None of these

Ans. d

Q14. [1 mark] Which of the following is a recurrence relation for the sequence 2, 3, 5, 8, 12, 17, 23, 30, ...? Assume that the sequence begins with the term called a_1.

(a) $a_n = a_{n-1} + (n-1)$
(b) $a_n = 2a_{n-1} - 1$
(c) $a_n = a_{n-1} + a_{n-2}$
(d) $a_n = a_{n-1} + n$

Ans. a

Q15. [1 mark] Which of the following is the negation of "Some bananas are ripe"?

(a) Some bananas are not ripe (b) No bananas are ripe (c) All bananas are ripe

Ans. b

Q16. [1 mark] Assume that you have an ordinary deck of 52 playing cards. How many possible 7-card poker hands are there that contain exactly one ace and one king?

(a) $C(4,1) \cdot C(4,1) \cdot C(44,5)$
(b) $C(8,2) \cdot C(44,5)$
(c) $52 \cdot 48 \cdot (44 \cdot 43 \cdot 42 \cdot 41 \cdot 40)$
(d) $C(52,7) - C(44,7)$

Ans. a

Q17. [1 mark] How many different license plates are available if the license plate pattern consists of four letters followed by three digits? (Assume that all letters are uppercase and the digits are 0, 1, ..., 9.)

(a) $26 \cdot 25 \cdot 24 \cdot 23 \cdot 10 \cdot 9 \cdot 8$
(b) $26^4 \cdot 10 \cdot 9 \cdot 8$
(c) $26^4 + 10^3$
(d) $26^4 \cdot 10^3$
(e) $C(26,4) \cdot C(10,3)$

Ans. d

Q18. [1 mark] How many different ways are there to select a committee consisting of three members of the U. S. Senate and three members of the U. S. House of Representatives? (There are 100 members of the U. S. Senate and 435 members of the U. S. House of Representatives.)

(a) $C(435,3) \cdot C(100,3)$
(b) $C(435,3) + C(100,3)$
(c) $C(535,6)$
(d) $435 \cdot 434 \cdot 433 \cdot 100 \cdot 99 \cdot 98$
(e) $435 \cdot 434 \cdot 433 + 100 \cdot 99 \cdot 98$

Ans. a

Q19. [1 mark] A dormitory has 40 students – 12 sophomores, 8 juniors, and 20 seniors. Which of the following is equal to the number of ways to form a committee of five students from the same class?

(a) $\frac{C(40,5)}{3}$
(b) $P(12,5) + P(8,5) + P(20,5)$
(c) $P(12,5) \cdot P(8,5) \cdot P(20,5)$
(d) $C(12,5) + C(8,5) + C(20,5)$
(e) $C(12,5) \cdot C(8,5) \cdot C(20,5)$

Ans. d

Q20. [2 marks] What is the power set of the set $\{\phi, \{0\}\}$?

Ans. $\{\phi, \{\phi\}, \{\{0\}\}, \{\phi, \{0\}\}\}$

Q21. [2 marks] Determine if the following describes a function with the given domain and codomain.

$$f: \mathbb{N} \to \mathbb{N}, \text{ where } f(n) = \begin{cases} n+4, \text{if } n < 7 \\ n^2, \text{if } n > 11 \end{cases}$$

Ans. This is not a function because $f(7)$ is not defined (neither case covers the values $n = 7,8,9,10,11$).

Q22. [2 marks] Suppose that the characteristic equation of a linear homogeneous recurrence relation with constant coefficients is: $(r-3)^4(r-2)^3(r+6) = 0$. Write the general solution of the recurrence relation.

Ans. $a_n = a3^n + bn3^n + cn^2 3^n + dn^3 3^n + e2^n + fn2^n + gn^2 2^n + h(-6)^n$

Q23. [2 marks] Let $f: \mathbb{R} \to \mathbb{R}$ have the rule $f(x) = \lfloor 3x \rfloor - 1$. Find $f^{-1}(S)$, where $S = \{0\}$.

Ans. $f^{-1}(0)$ is equal to the set of all x such that $\lfloor 3x \rfloor - 1 = 0$, or $\lfloor 3x \rfloor = 1$. Any number $x \in \left[\frac{1}{3}, \frac{2}{3}\right)$ will work.

Q24. [2 marks] Find the number of permutations of 1,2,…,8 that begin with 52 or end with 327.

Ans.

Let A be the set of permutations of $1, 2, \ldots, 8$ that begin with 52 and let B be the set of permutations of $1, 2, \ldots, 8$ that end with 327. In this case $A \cap B = \emptyset$ because the digit 2 cannot occur in both the string 52 and 327. Therefore,

$$|A \cup B| = 6! + 5! = 720 + 120 = 840.$$

Q25. [2 marks] Find the number of permutations of all 26 letters of the alphabet that contain at least one of the words CAR, CARE, SCARE, SCARED.

Ans.
Let A, B, C, and D be the sets of permutations of the 26 letters of the alphabet that contain the words CAR, CARE, SCARE, and SCARED, respectively. Then $D \subseteq C \subseteq B \subseteq A$. Hence,

$$|A \cup B \cup C \cup D| = |A| = 24!.$$

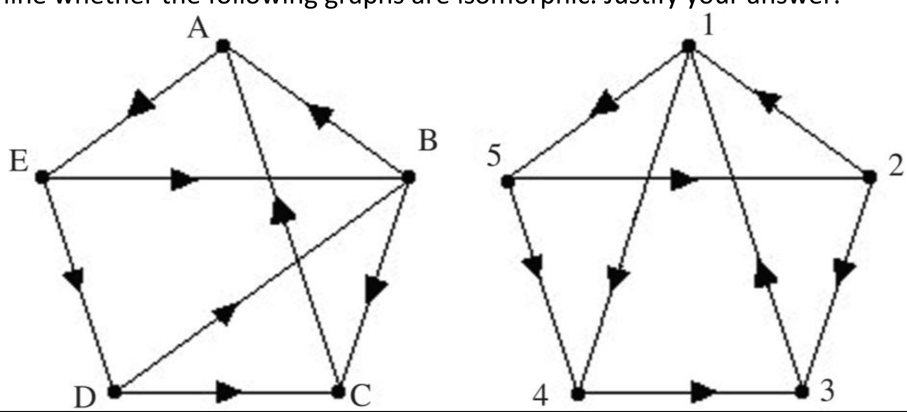
Q26. [2 marks] Find the coefficient of $a^{17}b^{23}$ in the expansion of $(3a - 7b)^{40}$.

Ans.
We expand $(3a-7b)^{40}$ using the binomial theorem, locate the term with the product $a^{17}b^{23}$, and then find the coefficient:

$$\begin{aligned}(3a-7b)^{40} &= \cdots + \binom{40}{17}(3a)^{17}(-7b)^{23} + \cdots \\ &= \cdots + \binom{40}{17} 3^{17}(-7)^{23}a^{17}b^{23} + \cdots.\end{aligned}$$

Thus, the coefficient is $\binom{40}{17} 3^{17}(-7)^{23}$, which can also be written as $\binom{40}{23} 3^{17}(-7)^{23}$.

Q27. [2 marks] Determine whether the following graphs are isomorphic. Justify your answer.



Ans. Even though the graphs have many features in common (such as the same number of vertices, the same number of edges, matching in-degrees and out-degrees), the digraphs are not isomorphic.

Here is one reason: Vertex B must correspond to vertex 1 because they are the only vertices with in-degree 2 and outdegree 2. Vertices D and E each have in-degree 1 and out-degree 2. If the two graphs are to be isomorphic, then D and E must correspond to 2 and 5 (in some order). Because there is an edge from E to D, there must be a corresponding edge in the digraph on the right—this forces D to correspond to 2 and E to correspond to 5. However in the left graph there is an edge from E to B, but no edge from 5 to 1 (the vertices corresponding to B and E) in the right graph. Therefore, the two digraphs are not isomorphic.

Q28. [2 marks] Find all solutions to $\lfloor x \rfloor \lceil x \rceil = x^2$.

Ans.

We first observe that every integer is a solution because in this case $\lfloor x \rfloor = x$ and $\lceil x \rceil = x$.

Now suppose that x is not an integer. Therefore, there is an integer n such that $\lfloor x \rfloor = n$ and $\lceil x \rceil = n + 1$.

Hence, in this case the original equation becomes $n(n+1) = x^2$, or $x = \pm \overline{n(n+1)} = \pm\sqrt{2}, \pm\sqrt{6}, \pm\sqrt{12}, \pm\sqrt{20}$, etc. Therefore, the solutions to the equation $\lfloor x \rfloor \lceil x \rceil = x^2$ are all integers and all numbers of the form $\pm\, n(n+1)$.

Q29. [2 marks] Determine whether the following is true for all 2×2 matrices: $(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + \mathbf{2AB} + \mathbf{B}^2$

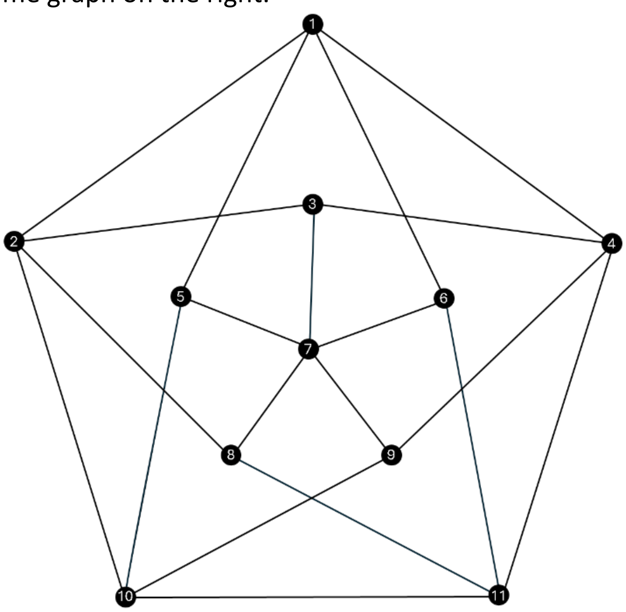
Ans.

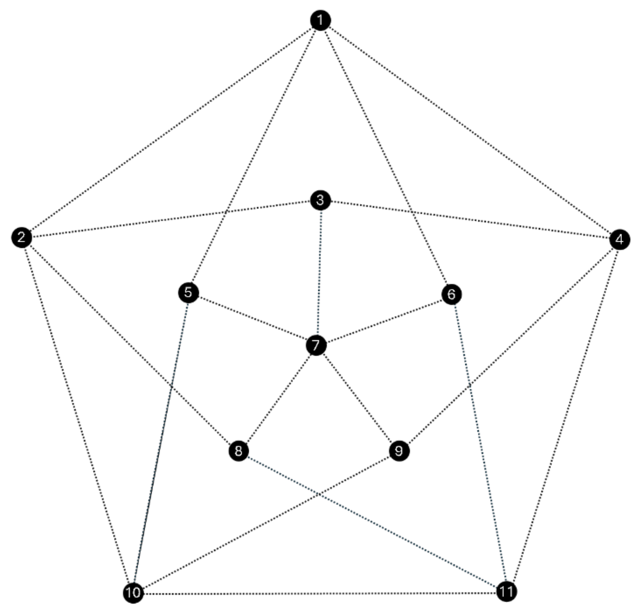
Using the distributive law to multiply the left side yields $(\mathbf{A}+\mathbf{B})^2 = (\mathbf{A}+\mathbf{B})(\mathbf{A}+\mathbf{B}) = \mathbf{A}^2+\mathbf{AB}+\mathbf{BA}+\mathbf{B}^2$. Therefore, the original equation is true if and only if $\mathbf{BA} = \mathbf{AB}$.

But $\mathbf{AB} \neq \mathbf{BA}$. For example, we can take $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then $\mathbf{AB} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ but $\mathbf{BA} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$.

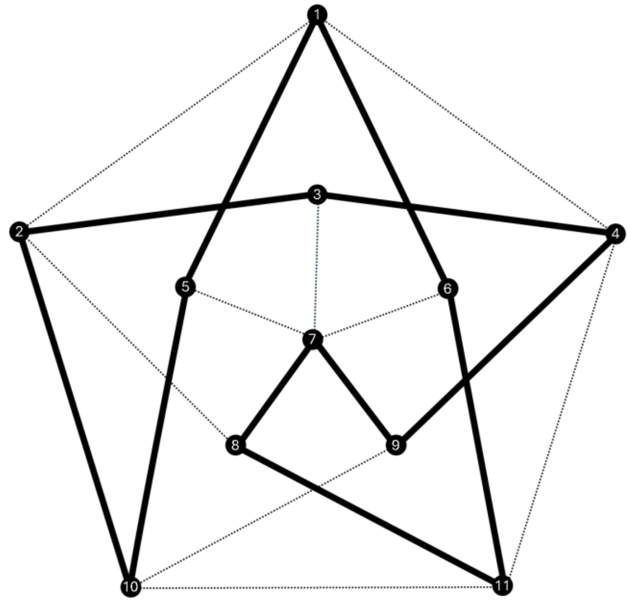
Therefore, the original equation is not always true.

Q30. [3 marks] For the following graph on the left, draw the edges of its Hamilton circuit using your pen in the copy of the same graph on the right.





Ans. There are multiple Hamilton circuits, one of which is displayed below.



Q31. [3 marks] Suppose you have a group of n people $(n \geq 2)$. Use the Pigeonhole Principle to prove that there are at least two in the group with the same number of friends in the group.

Ans.

Suppose you have a group of n people, and suppose that no two persons have the same number of friends in the group. There are n numbers of possible friends that a person can have: $0, 1, 2, \ldots, n-1$, where 0 means that the person has no friends in the group, 1 means that the person has exactly one friend in the group, …, $n-1$ means that the person is friends with all other people in the group.

It is impossible for the numbers 0 and $n-1$ to both occur as "friendship numbers" in a group of n people. To see this, suppose person A has 0 friends and person B has $n-1$ friends. If the friendship number for B is $n-1$, then B must be friends with everyone, including A. But this contradicts the fact that the friendship number for A is 0. Therefore, of the n possible friendship numbers, only $n-1$ friendship numbers are available in any group of n people.

Because there are n people in the group and only $n-1$ available friendship numbers, by the Pigeonhole Principle at least two people have the same friendship number. That is, at least two people in the group have the same number of friends in the group.

Q32. [3 marks] Find the number of words of length 10 of letters of the alphabet, with no repeated letters, such that each word has equal numbers of vowels and consonants.

Ans.

Visualize a row of ten blanks. Each word of length 10 must contain all five vowels and five of the 21 consonants. There are different ways to solve this problem.

Here is one way to solve the problem. Choose a set of five of the ten blanks for the vowels. Then place the five vowels in these positions. These two steps can be done in $C(10,5) \cdot 5!$ ways. Next, choose 5 consonants from the 21 consonants, and place them in the remaining five blanks. These two steps can be done in $C(21,5) \cdot 5!$ ways. Thus, the number of ways of forming the desired words by placing the vowels and consonants is

$$C(10,5) \cdot 5! \cdot C(21,5) \cdot 5! = \frac{10!}{5!5!} \cdot 5! \cdot \frac{21!}{5!16!} \cdot 5! = \frac{10!21!}{5!16!} = 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 21 \cdot 20 \cdot 19 \cdot 18 \cdot 17 = 73,842,451,200.$$

Here is a second way to approach the problem. Choose any set of five consonants, and form a set consisting of these five consonants and the five vowels. These ten letters can be arranged in 10! ways, each of which forms one of the words we are counting. But there are $C(21,5)$ ways to choose the five consonants. Therefore, there are $C(21,5)\cdot 10! = 10\cdot 9\cdot 8\cdot 7\cdot 6\cdot 21\cdot 20\cdot 19\cdot 18\cdot 17$ words.

Here is a third way to approach the problem. First, place the five vowels in five of the ten blanks (10 choices for A, 9 choices for E, etc.)—this can be done in $10\cdot 9\cdot 8\cdot 7\cdot 6$ ways. Then place five of the 21 consonants in the five remaining blanks (21 choices for the first blank, 20 choices for the second blank, etc.)—this can be done in $21\cdot 20\cdot 19\cdot 18\cdot 17$ ways. Therefore, by the product rule, the number of words with the five vowels and five consonants is $10\cdot 9\cdot 8\cdot 7\cdot 6\cdot 21\cdot 20\cdot 19\cdot 18\cdot 17$.

Q33. [4 marks] In how many ways can ten books be put in four labeled boxes, if one or more of the boxes can be empty? Assume that the books are:

(a) distinct
(b) identical

Ans.

(a) Each book can be placed in any of the four boxes. By the product rule, this can be done in 4^{10} ways.
(b) Let x_i be the number of books in box i. We need to find the number of nonnegative integer solutions to the equation $x_1 + x_2 + x_3 + x_4 = 10$. There are $C(13,3)$ such solutions.

Q34. [4 marks] For the sequence of Fibonacci numbers $f_0, f_1, f_2, \ldots$ (i.e., $0,1,1,2,3,5,8,13,\ldots$), prove for all nonnegative integers n: $f_0^2 + f_1^2 + f_2^2 + \cdots + f_n^2 = f_n f_{n+1}$.

Ans.

Let $P(n)$ be the proposition

$$f_0^2 + f_1^2 + f_2^2 + \cdots + f_n^2 = f_n f_{n+1}.$$

BASIS STEP: $P(0)$ is the proposition $f_0^2 = f_0 f_1$. It is true because $f_0^2 = 0^2 = 0$ and $f_0 f_1 = 0\cdot 1 = 0$.

INDUCTIVE STEP: Suppose $P(k)$ is true. Then

$$f_0^2 + f_1^2 + f_2^2 + \cdots + f_k^2 = f_k f_{k+1}.$$

We need to show that $P(k+1)$ is true: $f_0^2 + f_1^2 + f_2^2 + \cdots + f_{k+1}^2 = f_{k+1} f_{k+2}$. We take $P(k)$ and add f_{k+1}^2 to both sides of the equation, obtaining

$$\begin{aligned}(f_0^2 + f_1^2 + f_2^2 + \cdots + f_k^2) + f_{k+1}^2 &= f_k f_{k+1} + f_{k+1}^2 \\ &= f_{k+1}(f_k + f_{k+1}) \\ &= f_{k+1} f_{k+2} \qquad \text{(using the Fibonacci sequence recurrence)}\end{aligned}$$

Therefore $P(k+1)$ follows from $P(k)$.

Therefore, by the Principle of Mathematical Induction, $P(n)$ is true for all nonnegative integers n.

Q35. [4 marks] A bakery sells four kinds of cookies: chocolate, jelly, sugar, and peanut butter. You want to buy a bag of 30 cookies. Assuming that the bakery has at least 30 of each kind of cookie, how many bags of 30 cookies could you buy if you must choose:

(a) at least 3 chocolate cookies and at least 6 peanut butter cookies
(b) at most 5 sugar cookies

Ans.

We will use c to represent the number of chocolate cookies purchased, j for the number of jelly cookies purchased, s for the number of sugar cookies purchased, and p for the number of peanut butter cookies purchased.

(a) We must have $c \geq 3$ and $p \geq 6$. Picture yourself in the bakery with an empty bag in which you will place 30 cookies. Put into the bag three chocolate cookies and six peanut butter cookies. The bag now has nine cookies in it, so you need to place 21 more cookies into the bag. You have met the two conditions, so you do not care about the numbers of each of the four types you choose for the remaining 21 cookies. Therefore, you need to find the number of nonnegative integer solutions to the equation

$$c + j + s + p = 21.$$

The number of nonnegative integer solutions to this equation is $C(24, 3)$. Therefore there are $C(24, 3) = 2{,}024$ ways to buy 30 cookies with at least 3 chocolate cookies and at least 6 peanut butter cookies.

(b) To solve the problem, we use "complement counting". That is, we find the number of possible bags of 30 cookies with more than 5 sugar cookies, and then subtract that number from the total number of bags of 30 cookies. That is, we find the number of solutions to $c + j + s + p = 30$ with $s \geq 6$. This is the number of solutions to the equation $c + j + s + p = 24$, which equals $C(27, 3)$.

Thus, the answer to the question is the total number of nonnegative integer solutions to the equation $c + j + s + p = 30$, $C(33, 3)$, with $C(27, 3)$ subtracted:

$$(\text{total number of solutions}) - (\text{number of solutions with } s \geq 6) = C(33, 3) - C(27, 3) = 2{,}531.$$

Q36. [4 marks] You have two distinct parallel lines L_1 and L_2. You keep adding additional lines, $L_3, L_4, \ldots$ with none parallel to L_1 or L_2 or to each other, and no three passing through the same point.

(a) Find a recurrence relation and initial condition(s) for r_n, which is defined to be the number of regions into which the plane is divided by the lines $L_1, L_2, \ldots, L_n$.

(b) Find a closed-form formula for the number of regions into which the plane is divided by $L_1, L_2, \ldots, L_n$.

Ans.

(a) The recurrence relation is $r_n = r_{n-1} + n (n > 2)$, with initial condition $r_2 = 3$. To see this, note that line L_n must cut each of the previous $n - 1$ lines in exactly one point. This in effect divides L_n into n segments, each of which divides an existing region into two parts. Therefore, $r_n = r_{n-1} + n$.

(b) Proceeding inductively:

$$\begin{aligned} r_2 &= 3, \\ r_3 &= r_2 + 3 = 3 + 3, \\ r_4 &= r_3 + 4 = 3 + 3 + 4, \\ r_5 &= r_4 + 5 = 3 + 3 + 4 + 5, \\ r_6 &= r_5 + 6 = 3 + 3 + 4 + 5 + 6. \end{aligned}$$

This suggests that

$$\begin{aligned} r_n &= 3 + (3 + 4 + \cdots + n) \\ &= (1 + 2) + (3 + 4 + \cdots + n) \\ &= 1 + 2 + 3 + \cdots + n \\ &= \frac{n(n+1)}{2}. \end{aligned}$$

To verify that this guess is correct, take $r_n = r_{n-1} + n$ and substitute $\frac{n(n+1)}{2}$ for r_n and $\frac{(n-1)n}{2}$ for r_{n-1}, obtaining $\frac{n(n+1)}{2} = \frac{(n-1)n}{2} + n$, which is true because the right side simplifies to give the left side: $\frac{(n-1)n}{2} + n = \frac{(n-1)n + 2n}{2} = \frac{n^2 + n}{2} = \frac{n(n+1)}{2}$.

Q37. [4 marks] Solve: $a_n = -7a_{n-1} - 10a_{n-2}, a_0 = a_1 = 3$.

Ans.

Using $a_n = r^n$ yields the characteristic equation $r^2 + 7r + 10 = 0$, or $(r + 5)(r + 2) = 0$. Therefore the general solution is

$$a_n = c(-5)^n + d(-2)^n.$$

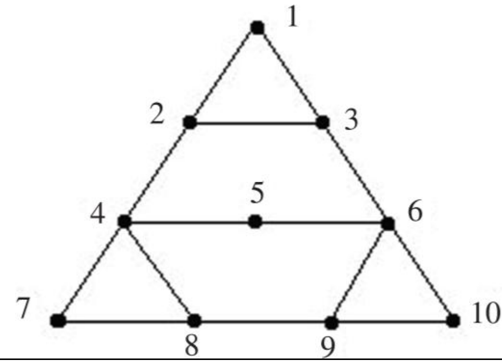
The initial conditions give the system of equations

$$\begin{aligned} c + d &= 3 \\ -5c - 2d &= 3. \end{aligned}$$

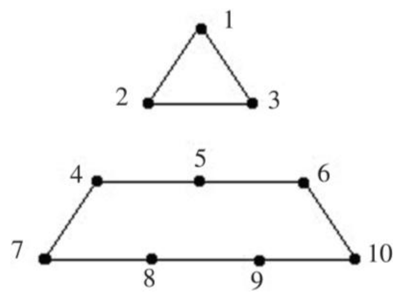
The solution to the system is $c = -3$ and $d = 6$. Hence, the solution to the recurrence relation is

$$a_n = (-3)(-5)^n + 6(-2)^n.$$

Q38. [4 marks] Determine whether the following graph has an Euler circuit, Euler path, Hamilton circuit and Hamilton path. Justify your answer.



Ans. The graph has no Euler circuit or path because it has 4 vertices of odd degree.
The graph does not have a Hamilton circuit. Suppose the graph did have a Hamilton circuit. Then the following edges must all be used in such a circuit: {1,2}, {1,3}, {4,5}, {5,6}, {4,7}, {7,8}, {6,10}, {9,10}. If these edges must be used, then the following edges cannot be used: {2,4}, {3,6}, {4,8}, and {6,9}. If it is impossible to use these four edges, they can be removed from the graph, yielding



But this graph has no Hamilton circuit because it is disconnected. Therefore, the original graph has no Hamilton circuit.
The graph does have a Hamilton path — for example, 1, 2, 3, 6, 5, 4, 7, 8, 9, 10.

Q39. [6 marks] Solve the recurrence relation $a_n = 8a_{n-1} - 12a_{n-2} + 3n$, with initial conditions $a_0 = 1$ and $a_1 = 5$.

Ans.

The characteristic equation for the associated homogeneous recurrence relation is $r^2 - 8r + 12 = 0$, which has solutions $r = 6$ and $r = 2$. Therefore, the general solution to the associated homogeneous recurrence relation is $a_n = a \cdot 6^n + b \cdot 2^n$. To obtain a particular solution to the given recurrence relation, try $a_n^{(p)} = cn + d$, obtaining

$$cn + d = 8[c(n-1) + d] - 12[c(n-2) + d] + 3n,$$

which can be rewritten as

$$n(c - 8c + 12c - 3) + (d + 8c - 8d - 24c + 12d) = 0.$$

The coefficient of n-term and the constant term must each equal 0. Therefore, we have

$$\begin{aligned} c - 8c + 12c - 3 &= 0 \\ d + 8c - 8d - 24c + 12d &= 0, \end{aligned}$$

or $c = 3/5$ and $d = 48/25$.

Therefore,

$$a_n = a6^n + b2^n + \frac{3}{5}n + \frac{48}{25}.$$

Using the two initial conditions, $a_0 = 1$ and $a_1 = 5$, yields the system of equations

$$\begin{aligned} a6^0 + b2^0 + \frac{3}{5} \cdot 0 + \frac{48}{25} &= 1 \\ a6^1 + b2^1 + \frac{3}{5} \cdot 1 + \frac{48}{25} &= 5 \end{aligned}$$

and the solution is found to be $a = 27/25$ and $b = -2$. Therefore, the solution to the given recurrence relation is

$$a_n = \frac{27}{25}6^n - 2^{n+1} + \frac{3}{5}n + \frac{48}{25}.$$